C U R R I C U L U M   V I T A E

Rahul Rawat

Master Thesis, AI Suitability Evaluation for Modelica Physical Models

PERSONAL DETAILS

Address	C2 16, 68159 Mannheim, Germany
Phone	015563603340
Email	rahulrawat2r@gmail.com
LinkedIn	linkedin.com/in/rahulrawat2r
GitHub	github.com/rahulrawat20022002
Portfolio	rah-portfolio.pages.dev
Date of birth	20 February 2002
Nationality	Indian, student visa with valid work permit
Availability	Werkstudent 20 hours per week immediately, full time from April 2027

PROFILE

Data Science and Analytics Master student at SRH Heidelberg based in Mannheim, with hands on evaluation of open weight and closed source large language models, honest benchmarking of alternative models on real data, and end to end machine learning delivery. I have shipped a modular Retrieval Augmented Generation system on Llama **3.1 8b** via Groq with a custom decision making router, a fairness by design credit scoring system with SHAP driven subgroup analysis and full regulatory documentation, and a recursive time series pipeline at eRay GmbH benchmarking six models head to head with strict anti leakage rules. Proficient in Python and comfortable with LLM APIs, prompt engineering, and structured evaluation harnesses, I am the right fit for the Master Thesis on AI Suitability Evaluation for Modelica Physical Models at Airbus Hamburg.

PROFESSIONAL EXPERIENCE

Oct 2025 to Mar 2026
Heidelberg

eRay GmbH

Data Scientist

- During a six month eRay GmbH and SRH Heidelberg collaboration to forecast water quality on a German lake, tasked with covering four indicators over rolling horizons, built an end to end recursive time series pipeline for chlorophyll a, turbidity, pH, and dissolved oxygen, delivered as a production ready module the client can rerun on every new sensor drop.
- Faced with model choice ambiguity for the forecasting task and the need to convey uncertainty to non technical stakeholders, benchmarked six candidates head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, then landed on CatBoost multi quantile regression, which produced asymmetric **80 percent** prediction intervals that gave the client decision support under uncertainty.
- Concerned that naive time series splits would inflate accuracy, tasked with making the evaluation defensible, enforced strict anti leakage rules across the pipeline, which surfaced the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors.
- With winter readings missing and tree based models flatlining during recursive prediction, needing realistic seasonal shape in the downstream forecasts, reconstructed missing values with MICE imputation and engineered a synthetic winter decay forecast canvas, which restored believable winter behaviour without introducing bias into the training window.
- To prevent bad data cascading downstream through the recursive forecaster, tasked with hardening the run loop, wrapped the pipeline in an orchestrator with gate checks and velocity and ecological bounds, so a failed imputation now halts the run rather than corrupting weeks of downstream predictions.

Technologies: *Python, CatBoost, Prophet, scikit learn, MICE*

Aug 2023 to Aug 2024
India

SS Engineers and Contractors

Junior Associate Software Developer

- With SS Engineers and Contractors running internal Data Dashboards, Analytics platforms, and Employee Portals used across the company, tasked with building and maintaining the front end features that day to day teams depended on, contributed React UI components across all three internal products, which stayed in daily use by internal teams across the company throughout the year in role.
- With a client relying on a legacy AngularJS app that had to sit inside an existing module federation setup alongside newer React micro frontends, tasked with porting the client's screens across without breaking the running product, ported around 8 routes from AngularJS to React inside the module federation shell over **4 months** of incremental releases, shipped with no production incidents during rollout.

Technologies: *React, module federation, Playwright, AngularJS, HTML5, CSS3*

Feb 2023 to Aug 2023
India

SS Engineers and Contractors

Front End Developer Intern

- During a six month internship at SS Engineers and Contractors, tasked with learning the codebase and then taking on small UI work across the internal Data Dashboards and Employee Portals under senior review, paired closely with senior developers to walk the codebase and then shipped focused UI components including charts, filters, and profile pages, iterating each one on code review feedback until it landed.

Technologies: *React, HTML5, CSS3, Git, code review workflow*

PERSONAL PROJECTS

2025

Hybrid RAG Orchestrator with Agentic Routing

- For a personal RAG project that needed to answer from local PDFs, live web, and small talk without a hardcoded path, tasked with building the router, delivered a modular Retrieval Augmented Generation system on top of Llama **3.1 8b** via Groq and LangChain orchestration with a custom decision making router that classifies user intent into three execution paths, local knowledge retrieval, external web search, or direct conversational logic, and returns clean end to end responses on all three routes.
- With document knowledge needing to persist across sessions rather than being reloaded each run, tasked with adding a durable semantic memory, built a ChromaDB store with local persistence and HuggingFace MiniLM L6 v2 embeddings over the PDF and vector data, which returned consistent retrieval on cross session queries.
- Because single shot responses were breaking follow up questions in testing, tasked with keeping multi turn context intact for the LLM, added a stateful MemoryAgent directly into the inference pipeline, which produced coherent follow up answers rather than isolated single shot responses.

Technologies: *Python, LangChain, Llama 3.1 8b via Groq, ChromaDB, HuggingFace MiniLM L6 v2, Streamlit*

2025

CreditIQ: Fairness by Design Credit Scoring

- For an SRH Heidelberg project on regulated credit scoring where a baseline model was failing the EU AI Act and AGG **80 percent** fairness bar, tasked with getting it back into compliance without gutting predictive quality, applied AIF360 mitigation and threshold calibration on a real credit dataset, which raised the Disparate Impact ratio from a failing **0.79** to a compliant **0.88**.
- After single axis age bias was fixed but younger women were still being penalised, tasked with finding and correcting the hidden intersectional bias, used SHAP driven subgroup analysis to expose the pattern and designed a four way age by gender threshold matrix, which corrected it without over correcting into reverse discrimination.
- With a large false negative rate silently rejecting good applicants, tasked with cutting it while keeping overall accuracy defensible, documented the fairness accuracy trade off as a deliberate and regulator defensible decision, which brought the false negative rate down from **44 percent** to **16.7 percent** while accuracy held at **75 percent** on the held out test split.

Technologies: *Python, scikit learn, AIF360, SHAP, Streamlit*

EDUCATION

Apr 2025 to Present
Heidelberg

M.Sc. Data Science and Analytics
SRH University of Applied Sciences Heidelberg, GPA 1.9

2019 to 2023
Greater Noida, India

Bachelor of Technology in Computer Science
GL Bajaj Institute of Technology and Management, CGPA 7.3 of 10

RESEARCH AND THESIS

2022 to 2023
Greater Noida, India

Bachelor Thesis, Diabetes Prediction Using Machine Learning
GL Bajaj Institute of Technology and Management, IEEE style paper

- For a Bachelor thesis on diabetes prediction with a small clinical dataset of **768 patients**, tasked with building a defensible model comparison the examiners could audit, built a full end to end machine learning pipeline comparing six classifiers with **10 fold** cross validation and per model confusion matrices, delivering a model comparison that stood up in the thesis defence.
- Spotting biologically impossible zero values in the source data that the original authors had overlooked, tasked with restoring data integrity before any model fit, applied IQR based outlier removal and proper imputation, which lifted the dataset from silently broken to a clean training input for every downstream model.
- With a **65 to 35** class imbalance that made accuracy a misleading headline metric, tasked with choosing an evaluation that would not hide errors on the minority class, moved the headline metric from accuracy to ROC AUC, which exposed the real error patterns the accuracy score had been masking and gave the thesis an honest performance comparison.
- With the results needing to be publishable in substance rather than just submissible, tasked with writing them up formally, produced an IEEE style paper including an honest limitations section and what to do differently in a follow up study, which the supervisor accepted as publishable in substance.

Technologies: *Python, scikit learn, Pandas, Seaborn*

CERTIFICATIONS

- NVIDIA: Building LLM Applications With Prompt Engineering, issued November 2025.
- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.
- Google Data Analytics: Foundations, Data, Data, Everywhere, issued April 2025 on Coursera.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity, and applied AI on real world challenges.

LANGUAGES

English	Fluent, written and spoken
German	B1 in progress toward B2
Hindi	Native